Echoes of Value: How Customer Patterns Predict Profits and What the Data Say Aloud*

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In the context of digital transformation and increasingly data-driven decision-making, a comprehensive understanding of customer behavior has emerged as a critical factor in enhancing corporate competitiveness. This study leverages a dataset curated by Dr. Omar Romero-Hernandez – Berkeley Haas School of Business and investigates six key business challenges: user engagement prediction, customer churn analysis, future purchase forecasting, identification of potentially interested users, high-value customer profiling, and detection of high-consumption potential segments. By integrating behavioral data, transactional records, and demographic indicators such as household structure and income, the research applies machine learning techniquesincluding logistic regression, decision trees, and random forestsfor predictive modeling. Additionally, K-Means clustering is employed to construct a customer value assessment framework based on the RFM (Recency, Frequency, Monetary) model. The results offer actionable insights for target customer identification, segmentation, and precision marketing, aiming to boost repurchase rates, improve retention, and optimize resource allocation. This methodology demonstrates strong generalizability and applicability across multiple industries including retail, e-commerce, and financial services, facilitating intelligent customer relationship management and personalized service delivery.

0 Executive Summary

This report presents an in-depth analysis of **user behavior** on the companys e-commerce mobile app and website platform, aiming to identify **customer structure**, **loyalty segmentation**, **consumption potential**, and **conversion inefficiencies**. Findings reveal that the core user base is predominantly composed of middle-aged to senior individuals (ages 4070), with a large proportion categorized as **loyal customers**, albeit with moderate to low spending power. The share of **VIP customers** is relatively small, and their spending does not significantly exceed the platform average. Despite a healthy influx of new and potential users, **customer churn** remains notably high. Moreover, a substantial number of web users engage in browsing activities without completing purchases, indicating friction in the **conversion funnel**. To address these issues, the following strategic initiatives are proposed: (1) Implement data-driven **customer segmentation models**, such as **RFM analysis** and **clustering**, to personalize marketing efforts for the high-engagement, mature demographic segment (Kotler et al., 2016); (2) Enhance **loyalty** and **high-value customer** management through tiered incentives, personalized rewards, and **predictive modeling** to identify and nurture potential VIPs (Reinartz & Kumar, 2002); (3) Optimize the websites **conversion rate** by improving **user experience**, reducing friction points, and incorporating behavioral triggers such as urgency-based promotions and personalized offers (Brynjolfsson & Smith, 2000); and (4) Establish a **churn prediction framework** using behavioral indicators and deploy automated retention campaigns targeting at-risk users (Verbeke et al., 2012). These strategies collectively aim to improve **customer retention**, increase **conversion efficiency**, and enhance the **lifetime value** of key user segments. Supporting literature includes Kotler et al. (2016), Reinartz & Kumar (2002), Brynjolfsson & Smith (2000), Verbeke et al. (2012), and Kim & Mauborgne (2005), which provide theoretical and practical foundations for **customer-centric digital marketing** and **strategic growth**.

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1 Introduction

The company under analysis operates in the highly competitive e-commerce industry, offering both a mobile application and a website platform for digital retail transactions. As a medium-sized enterprise with a growing customer base, it has established a solid foothold in the online retail sector, particularly among middle-aged and senior consumers. Despite its stable presence, the company is currently facing critical strategic and operational challenges that could hinder its long-term growth.

1.1 Main Problems

Recent user behavior data reveal a complex picture: while the platform attracts a consistent flow of new and returning users, the core customer base is dominated by individuals aged 4070, many of whom demonstrate high levels of engagement and loyalty, yet exhibit moderate to low purchasing power. Furthermore, although the platform has a segment of VIP customers, their contribution to overall revenue remains limited, and their spending does not significantly exceed the average. Customer churn is also a significant issue, especially among new users, and a large proportion of web traffic fails to convert into purchases.

1.2 Business Questions

Against this backdrop, this report aims to provide a data-driven analysis of user behavior, focusing on four key areas: (1) customer structure, (2) loyalty segmentation, (3) consumption potential, and (4) conversion inefficiencies. The primary business questions guiding this analysis include:

- What are the demographic and behavioral characteristics of the platforms core and high-potential customers?
- How can the company improve customer segmentation and better target high-value users?
- What factors contribute to customer churn and low conversion rates, and how can they be mitigated?
- Which strategic initiatives can enhance retention, conversion, and lifetime value?

1.3 Analysis Objectives of Report

This report proposes a set of actionable strategies grounded in academic literature and business practice: the application of RFM analysis and clustering techniques for segmentation (Kotler et al., 2016), loyalty program optimization and predictive modeling for VIP nurturing (Reinartz & Kumar, 2002), UX and funnel optimization using behavioral triggers (Brynjolfsson & Smith, 2000), and the development of a churn prediction framework (Verbeke et al., 2012). Together, these approaches aim to strengthen the companys customer-centric digital strategy.

2 Methodology

The company under analysis operates in the highly competitive e-commerce industry, offering both a mobile application and a website platform for digital retail transactions. As a medium-sized enterprise with a growing customer base, it has established a solid foothold in the online retail sector, particularly among middle-aged and senior consumers. Despite its stable presence, the company is currently facing critical strategic and operational challenges that could hinder its long-term growth.

2.1 Data Source

The dataset utilized in this analysis is provided by Dr. Omar Romero-Hernandez from the Berkeley Haas School of Business.magentahttps://haas.berkeley.edu/faculty/romerohernandez-omar/ It comprises comprehensive customer data collected from the companys e-commerce platform, capturing multiple dimensions of user behavior, demographics, and transactional history. The data enables a holistic understanding of customer profiles and purchasing patterns, essential for strategic business analysis and decision-making. Data path magentahttps://www.kaggle.com/datasets/imakash3011/customer-personality-analysis/data

Data Size:

$$X \in R^{n \times p}, \quad \text{where } n = 2216 \text{ samples } p = 29 \text{ cols}(0)$$

2.2 Data Preview

The dataset includes the following categories and key attributes:
red* **Customer Demographics and Profile:** Each customer is uniquely identified by an ID and associated with demographic information such as year of birth, education level, marital status, household income, and family composition.

Table 1. Customer Demographics and Profile

ID	Year Birth	Education	Kid Home
5524	1957	Graduation(2)	0(No kids)
5324	1981	PhD(4)	1(Has kids)
8180	1952	Master(3)	1(Has kids)

ID	Teenhome	Income	Marital Status
5524	0 (No teens)	$58,138	Single(3)
5324	0 (No teens)	$58,293	Married(1)
8180	1 (Has teens)	$59354	Divorced(4)

red* **Transactional and Consumption Data:** Detailed records of customer spending in various product categories over the past two years are provided, including expenditures on wine, fruits, meat, fish, sweets, and gold products.

Table 2. Customers Transactional Consumption Data

ID	MntWines	MntFruits	MntMeat
5524	$635	$88	$546
5324	$173	$43	$118
8180	$233	$2	$53

ID	MntFish	MntSweets	MntGold
5524	$172	$88	$88
5324	$46	$27	$15
8180	$3	$5	$14

red* **Promotional Response and Engagement:** The dataset tracks customers interactions with marketing campaigns, including the number of purchases made with discounts, and binary indicators for acceptance of offers across multiple campaigns. It also includes responses to the most recent campaign, allowing assessment of promotional effectiveness.

Table 3. Customers Promotion Response and Engagement

ID	NumDeal	NumWeb	Response
5524	3	8	1
2174	2	1	0
4141	1	8	0

red* **Customer Experience Indicators:** Data on customer complaints within the last two years is included as a measure of customer satisfaction and potential churn risk.

Table 4. Customer Complaint Indicator (Last Two Years)

ID	5524	2174	4141	6182
Complaint Posted	0	0	0	0

This multi-dimensional dataset serves as the foundation for advanced customer behavior analysis, enabling the identification of core customer segments, loyalty patterns, consumption potential, and inefficiencies in the conversion funnel. It also supports the development of predictive models for retention and targeted marketing strategies, thereby informing the companys efforts to enhance customer lifetime value and operational performance.

Acknowledgement: The authors gratefully acknowledge Dr. Omar Romero-Hernandez and the Berkeley Haas School of Business for providing access to this valuable dataset.

2.3 Visualization

2.3.1 All-Features Visualization

In this analysis, we conducted a systematic visualization exploration of the multidimensional features contained in the data set redFigure 1, covering both **qualitative and quantitative variables**. For **qualitative variables** (such as the degree of user participation in the company's marketing activities, marital

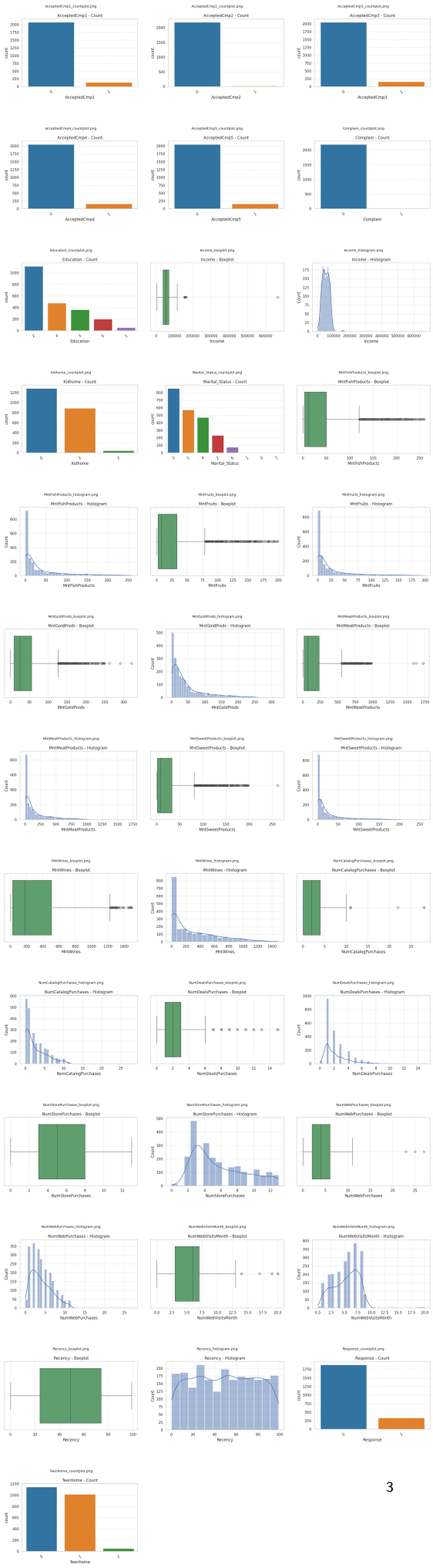


Fig. 1. Visualization of All Features

status, education level, etc.), we used frequency statistics and other methods (see sub-Figure 1-5) to display their distribution characteristics to identify user behavior preferences and distribution differences between categories. For **quantitative variables** (such as income, the amount of money spent on a specific commodity category, the number of children, etc.), box plots are mainly used to characterize their value range, central trend, and outliers. Above visualization methods, we not only clarified the **distribution range and concentration area** of various feature values, but also gained some preliminary insights. For example, from the user age distribution map, it can be observed that the age of registered users on the platform is mainly concentrated between 40 and 70 years old, indicating that the middle-aged and elderly groups are the core user population of this platform. These analyses provide an important basis for subsequent user segmentation and strategy formulation.

2.3.2 Feature Correlation Matrix Visualization

In the context of high-dimensional datasets comprising a wide range of features such as consumer household demographics, income levels, behavioral responses to campaigns, and purchase historiesit is essential to identify and retain the most relevant variables to improve model efficiency, reduce overfitting, and enhance interpretability. To this end, we adopted a **correlation-based feature selection** approach by constructing a **feature correlation matrix**, which quantitatively evaluates the relationships between pairs of features.

In particular, we employed the **Pearson Correlation Coefficient** to measure the *linear dependency* between continuous variables. The Pearson correlation coefficient between two random variables X and Y, denoted as $\rho_{(X,Y)}$, is defined as follows:

$$\frac{\mathrm{Cov}(X,Y)}{\sigma_X \sigma_Y} = \frac{\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i-\bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i-\bar{y})^2}}$$

Where:

- $\mathrm{Cov}(X,Y)$ is the covariance between variables X and Y,
- σ_X and σ_Y denote the standard deviations of X and Y, respectively,
- $\bar{x}$ and $\bar{y}$ represent the means of X and Y,
- n is the number of samples.

The value of $\rho_{X,Y}$ lies in the range $[-1,1]$, where:

- $\rho = 1$: perfect positive linear correlation,
- $\rho = -1$: perfect negative linear correlation,
- $\rho = 0$: no linear correlation.

Based on the resulting correlation matrix, we visualized inter-feature dependencies using heatmaps and systematically examined pairs of highly correlated features (those with $|\rho| > S_j = \sum_{\substack{i=1 \\ i\neq j}}^{d} |\rho_{X_i,X_j}|$).

Table 5. Feature correlations with Response

Feature	Correlation	Above avg
Response	1.0000	1
AcceptedCmp5	0.3234	1
AcceptedCmp1	0.2973	1
TotalSpend	0.2641	1
AcceptedCmp3	0.2540	1
MntWines	0.2463	1
MntMeatProducts	0.2377	1
NumCatalogPurchases	0.2199	1
CustomerTenure	0.1965	1
AcceptedCmp4	0.1802	1
AcceptedCmp2	0.1693	1
NumWebPurchases	0.1514	1
MntGoldProds	0.1403	1
Income	0.1330	1
MntFruits	0.1224	1
MntSweetProducts	0.1162	1
MntFishProducts	0.1081	1
Education	0.0908	0
NumStorePurchases	0.0362	0
Year$_{B}irth$	0.0237	0
NumDealsPurchases	0.0035	0
Complain	-0.0020	0
NumWebVisitsMonth	-0.0022	0
Marital$_{S}tatus$	-0.0126	0
ID	-0.0215	0
Age	-0.0237	0
Kidhome	-0.0779	0
Teenhome	-0.1539	1
TotalChildren	-0.1676	1
Recency	-0.1998	1
Z_CostContact	NaN	Nan
Z_Revenue	NaN	Nan

Table above redTable 5 shows that high participation in activities (AcceptedCmp5, 0.3234; AcceptedCmp1, 0.2973) is positively correlated with participation. However, users with teenagers (Teenhome, -0.1539) in families have lower participation due to time constraints, which is negatively correlated. The visualization of matrix shown as fellow:redFigure 2

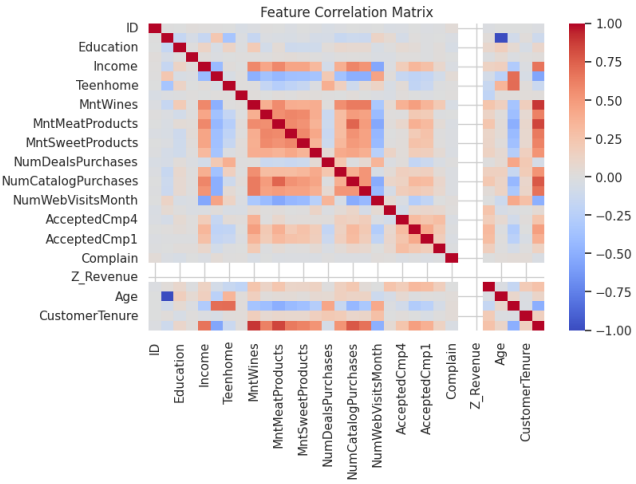


Fig. 2. Feature Correlation Matrix

2.4 Application of Models and Statistical Methods

To address the objectives of predicting user participation in the anniversary event, identifying potential churn, and assessing purchase intent, a suite of robust statistical and machine learning methodologies was employed. The following outlines the models and techniques utilized, emphasizing their methodological rigor and application, with key mathematics.

2.4.1 Classification Models for Predictive Tasks

For the binary classification task of predicting user participation in the anniversary event, as well as identifying at-risk customers and forecasting purchase intent, the following classical machine learning algorithms were applied:

- **Logistic Regression**: Utilized as a baseline model for its interpretability and effectiveness in binary classification tasks, leveraging a probabilistic framework. This formula predicts prob:

$$P(y = 1 \mid \mathbf{x}) = \frac{1}{1 + e^{-\mathbf{w}^T \mathbf{x} - b}} \tag{1}$$

where $\mathbf{x}$ is the feature vector, $\mathbf{w}$ is the weight vector, b is the bias, and $P(y = 1 \mid \mathbf{x})$ is the probability of the positive class, shown as redFigure 3

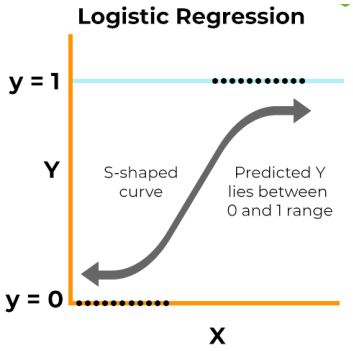


Fig. 3. Logistic Regression Probability Distribution

- **Decision Trees**: Employed to capture nonlinear relationships and feature interactions, providing a transparent decision-making structure shown as redFigure 4 The tree splits the feature space based on feature thresholds to minimize impurity, measured by the Gini index:

$$\text{Gini} = 1 - \sum_{i=1}^{C} p_i^2 \tag{2}$$

where p_i is the proportion of class i in a node, and C is the number of classes.

A decision tree is a hierarchical algorithm that recursively partitions the feature space into binary segments based on feature attributes, enabling predictive modeling of outcomes. In this application, the algorithm leverages binary (yes/no) responses derived from user information to construct a series of decision nodes.

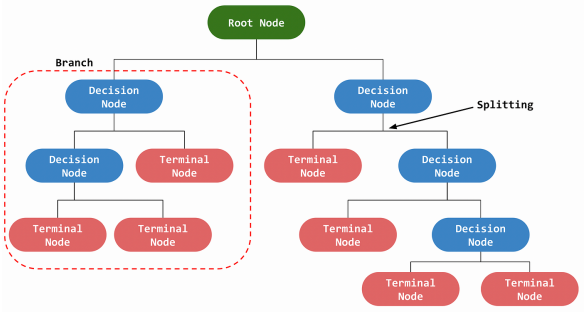


Fig. 4. Decision Tree Partition Structure

- **Random Forests**: Implemented to enhance predictive accuracy through ensemble learning, aggregating multiple decision trees. The final prediction is based on majority voting across T trees:

$$\hat{y} = \text{mode}\{h_1(\mathbf{x}), h_2(\mathbf{x}), \ldots, h_T(\mathbf{x})\} \tag{3}$$

where $h_t(\mathbf{x})$ is the prediction of the t-th tree.

These models were trained and validated to ensure reliable predictions, with performance metrics (e.g. confusion matrix) evaluated to select the optimal approach.

2.4.2 Multi-Dimension Statistic and Visualization

To characterize users with low purchase intent (e.g., those who browse without purchasing), a multidimensional statistical analysis was conducted:

- **Exploratory Data Analysis (EDA)**: Key user attributes were visualized to identify patterns and correlations, using statistical measures such as the correlation coefficient:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \tag{4}$$

where x_i and y_i are feature values, and $\bar{x}$, $\bar{y}$ are their means.

- **User Profiling**: Descriptive statistics, such as mean and variance, were applied to summarize user information:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i, \quad \sigma^2 = \frac{1}{n} \sum_{i=1}^{n} (x_i - \bar{x})^2 \tag{5}$$

The resulting profiles were compiled into structured outputs file.

2.4.3 Customer Segmentation via PCA and Cluster

To segment the customer base and identify high-value users, the following methodology was adopted:

- **Principal Component Analysis (PCA)**: Applied for dimensionality reduction to transform high-dim data into a lower-dim space. PCA identifies principal components by maximizing

variance, computed via eigenvalue decomposition of the covariance matrix:

$$\text{Cov}(\mathbf{X}) = \frac{1}{n-1}\mathbf{X}^T\mathbf{X}, \quad \mathbf{X} \text{ centered} \tag{6}$$

where $\mathbf{X}$ is the data matrix, and principal components are the eigenvectors corresponding to the largest eigenvalues.

- **Clustering Analysis**: Utilized to group customers into homogeneous clusters, using k-means clustering, which minimizes the within-cluster sum of squares:

$$J = \sum_{k=1}^{K} \sum_{\mathbf{x}_i \in C_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 \tag{7}$$

where $\boldsymbol{\mu}_k$ is the centroid of cluster C_k, and K is the number of clusters.

- **High-Value Customer Identification**: High-value groups were isolated based on cluster characteristics, with profiles recorded for strategic.

- **Statistical Distribution Analysis**: The distribution of customer types was quantified, often using frequency distributions or histograms to summarize cluster compositions.

This PCA-clustering approach ensured a rigorous segmentation process, enabling the prioritization of high-value customers and tailored engagement.

3 Company and Industry Overview

3.1 Company Overview

The company is dedicated to delivering premium, high-quality consumer goods that enhance customer lifestyles, fostering loyalty through personalized, omnichannel experiences and sustainable practices.

3.1.1 Mission

Provide better marketing services, product quality, and user experience.

3.1.2 Vision

To lead the global premium retail sector by offering innovative, customer-centric products and services, driven by data analytics and a commitment to sustainability.

3.1.3 Product Line

The company specializes in a diverse portfolio of premium consumer goods, including:

- **Beverages**: Premium wines, with an average customer spend of \$255.28, reaching up to \$1,032.
- **Food Products**: Fresh and processed items such as fruits (avg. \$19.96), meat (avg. \$164.75), fish (avg. \$37.66), and sweets (avg. \$31.03).
- **Specialty Goods**: Gold products (avg. \$40.66) and other luxury items tailored for affluent consumers.

Products are distributed through omnichannel platforms (web, catalog, in-store), with customers averaging 3.65 web purchases, 2.72 catalog purchases, and 5.24 store purchases.

3.1.4 Customer Base

The customer base comprises primarily affluent, educated individuals with the following characteristics:

- **Demographics**: Average age approximately 55 years (born 1943–1996), with 65% holding a Graduation degree, 22% PhD, 8% Master, 4% 2n Cycle, and 1% Basic education. Marital status varies (31% Married, 26% Together, 22% Single, 18% Divorced, 3% Widow).
- **Income**: Average income of \$49,614 (excluding missing values), ranging from \$2,447 to \$88,194, reflecting a mix of middle- and high-income consumers.
- **Purchasing Behavior**: High engagement in premium categories (e.g., meat and wines), with low deal-driven purchases (avg. 2.48). Customers average 5.82 web visits per month and show varied campaign responsiveness (12% accepted at least one campaign).
- **Loyalty**: 17% responded positively to marketing campaigns (Response = 1), with higher responsiveness among higher-income, older customers.

3.2 Industry Overview

The company operates in the premium consumer goods and retail industry, focusing on food, beverages, and specialty products.

3.2.1 Industry Structure

The industry is characterized as fellows:

- **Fragmented Market**: A mix of global chains, specialty retailers, and e-commerce platforms.
- **Omnichannel Dominance**: Integration of online (web/catalog) and physical retail to meet consumer demand for convenience.
- **Supply Chain Complexity**: Global sourcing for premium goods, requiring robust logistics and quality assurance.

3.2.2 Market Share

The company likely holds a niche position in the premium retail segment, targeting affluent consumers with high-quality products. Its omnichannel presence and data-driven marketing (12% campaign acceptance rate) provide a competitive edge, though it competes with larger players for broader market share.

3.2.3 Main Competitors

Key competitors include:

- **Mass Retailers**: Walmart, Costco, and Kroger, leveraging economies of scale.
- **Specialty Retail**: Whole Foods, Trader Joe's, and local gourmet stores, competing in premium niches.
- **E-commerce Platforms**: Amazon, Instacart, and direct-to-consumer brands, emphasizing convenience and subscriptions.
- **Luxury Goods Retailers**: Brands focusing on items like gold products or artisanal foods.

3.3 PESTEL Analysis

3.3.1 Political Factors

- **Trade Regulations**: Tariffs on imported goods (e.g., wines, fish) impact costs, particularly with trade tensions affecting specialty product sourcing.
- **Tax Policies**: Incentives for sustainable practices could benefit operations, while luxury goods taxes may affect pricing.
- **Political Stability**: Stable markets (e.g., U.S., EU) support operations, but geopolitical risks in sourcing regions could disrupt supply chains.

3.3.2 Economic Factors

- **Consumer Spending**: The average customer income ($49,614) supports demand for premium goods, but inflation may reduce discretionary spending.
- **Exchange Rates**: Currency fluctuations impact costs of imported goods like wines ($255.28 avg. spend).
- **Economic Trends**: Rising disposable incomes in developed markets bolster demand, but economic uncertainty may shift preferences toward budget alternatives.

3.3.3 Social Factors

- **Consumer Trends**: Demand for sustainable, ethically sourced products aligns with offerings (e.g., high meat/fish purchases). Educated customers (87% Graduation/PhD) prioritize quality.
- **Demographic Shifts**: Aging customers (avg. age 55) drive premium goods demand, while younger consumers seek digital engagement.
- **Lifestyle Preferences**: Single and divorced customers (40%) show higher campaign responsiveness, offering targeted marketing opportunities.

3.3.4 Technological Factors

- **E-commerce Growth**: High web purchases (avg. 3.65) and web visits (avg. 5.82/month) emphasize the need for digital platforms and AI-driven personalization.
- **Data Analytics**: Leveraging purchase data (e.g., 12% campaign acceptance) enables targeted marketing.
- **Supply Chain Tech**: Automation and blockchain enhance efficiency and transparency for premium goods.

3.3.5 Environmental Factors

- **Sustainability Expectations**: Customers demand eco-friendly packaging and sourcing for products like fish and meat.
- **Climate Risks**: Resource scarcity (e.g., fish, meat) may increase costs, impacting margins for high-spend categories.
- **Regulatory Pressure**: Stricter environmental laws require investments in green logistics.

3.3.6 Legal Factors

- **Consumer Protection**: Compliance with food safety and labeling regulations is critical for meat ($164.75 avg.) and fish ($37.66 avg.).
- **Data Privacy**: Adherence to GDPR/CCPA for customer data (e.g., purchase history) is essential.
- **Trade Compliance**: Import/export laws for specialty goods necessitate robust legal frameworks.

3.4 Overview

The company is strategically positioned in the premium retail market, serving an affluent, educated customer base with a diverse product line and omnichannel presence. Data insights (e.g., high wine/meat purchases, 17% campaign response rate) highlight opportunities for personalized marketing and loyalty programs. However, competition from mass retailers and e-commerce giants, combined with macroeconomic challenges, requires agility. The PESTEL analysis underscores the need to leverage technology, align with sustainability trends, and navigate regulatory complexities to maintain a competitive edge.

4 Internal Business Analysis

The internal statistic data shows that although the number of registered users on the platform has increased year by year, the company has recently discovered that some core operating problems have gradually emerged. In order to solve **1.2 Business Question** mentioned above, the company hopes to rely on historical transactions, behavior logs and customer attribute data to build a data-driven customer insight and prediction system. The company is now preparing to hold an anniversary celebration to activate user vitality and discover new user inflows, the tasks we need to solve are shown as fellows:

To enhance customer lifecycle management and data-driven decision-making, this study focuses on six key analytical tasks:

- **Task 1:** Predicting User Participation
- **Task 2:** Identifying Risk of Churn
- **Task 3:** Forecasting Purchase Intentions
- **Task 4:** Detecting Low-Conversion Users
- **Task 5:** Profiling High-Value Customers
- **Task 6:** Constructing a Customer Value Model and User Distribution Analysis

4.1 Task1 Analysis: Predicting User Participation

The company is preparing to launch an anniversary event. For this event, we need to **predict and estimate the user participation in advance**. For user information, we first need to pay attention to the **age** (shown as redFigure 5) distribution and **income** (shown as redFigure 6) of the user group, which directly affect consumption habits and the degree of response to the event. Next, we will visualize the distribution of these two important indicators for the degree of response:

4.1.1 Basic Visualization

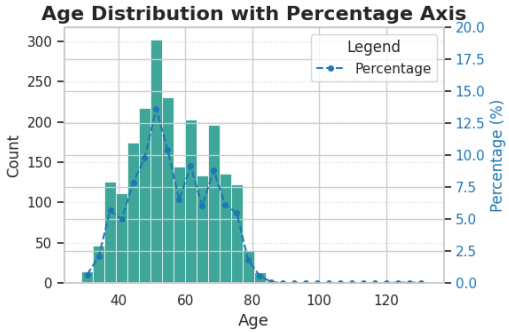


Fig. 5. Customer Age Distribution and Percentage

From the **distribution plot**, we can find that the above age distribution is centering on range between 40-70, which is **middle and old age**, then our company needs to formulate corresponding marketing strategies based on this specific age characteristic, and take care of them

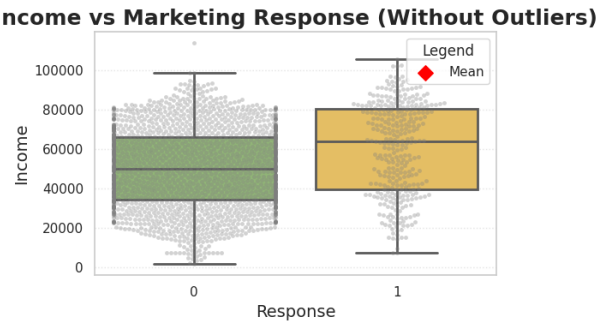


Fig. 6. Box Plot of Income and Responsiveness

From the **box plot**, we found that people with generally higher annual incomes are more motivated to participate in the response activities, but this part of the response population is not large.

4.1.2 Model Implementation

This is a **binary classification task**, denoted as $y \in \{0,1\}$, where $y = 1$ indicates participation and $y = 0$ indicates non-participation. We will employ three classic classification algorithms: **logistic regression, decision trees, and random forests** for prediction.

Let the original dataset be $D = \{(x_i, y_i)\}_{i=1}^{N}$, where $x_i \in R^d$ represents feature vector of the i-th user, and y_i is the corresponding label. The preprocessing steps include:

- **Data Cleaning**: Handle missing values, outliers, and standardize features (e.g., normalize feature x_j as $x_j \leftarrow \frac{x_j - \mu_j}{\sigma_j}$, where μ_j and σ_j are the mean and standard deviation of feature j).
- **Dataset Splitting**: Randomly divide the dataset D into a training set D_{train} (70%) and a testing set D_{test} (30%), satisfying $|D_{\text{train}}| \approx 0.7N \approx 1551$ and $|D_{\text{test}}| \approx 0.3N \approx 665$.

4.1.3 Classification Algorithms

- **Logistic Regression**: Logistic regression models the probability $P(y = 1|x) = \sigma(w^T x + b)$, where $\sigma(z) = \frac{1}{1+e^{-z}}$ is the sigmoid function. The loss function is:

$$L(w,b) = -\frac{1}{N}\sum_{i=1}^{N}\Big[y_i \log(\sigma(w^T x_i + b)) + (1 - y_i)\log(1 - \sigma(w^T x_i + b))\Big]$$

- **Decision Tree**: A decision tree splits the feature space using metrics like information gain:

$$\text{Gain}(j,s) = H(D) - \sum_{k\in\{\text{l,r}\}} \frac{|D_k|}{|D|} H(D_k)$$

- **Random Forest**: Combines predictions from T decision trees:

$$\hat{y} = \text{mode}\{h_t(x)\}_{t=1}^{T}$$

4.1.4 Evaluation Metrics

Model performance will be evaluated using accuracy, precision, recall, and F1 score. The F1 score is defined as:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

where $\text{Precision} = \frac{\text{TP}}{\text{TP+FP}}$ and $\text{Recall} = \frac{\text{TP}}{\text{TP+FN}}$. By implementing and comparing logistic regression, decision trees, and random forests, we will select the optimal model for predicting user participation based on performance on the test set. Among the evaluated models, the logistic regression model demonstrated the most robust over-

Table 6. Task1 Classification Model Comparision

Indicator	LR		DT		RF	
	0	1	0	1	0	1
Precision	0.87	0.62	0.88	0.49	0.86	0.50
Recall	0.98	0.15	0.95	0.30	0.98	0.10
F1-Score	0.92	0.24	0.91	0.37	0.92	0.17
Accuracy	0.86		0.85		0.85	
Macro F1	0.58		0.64		0.54	
Weighted F1	0.82		0.83		0.80	

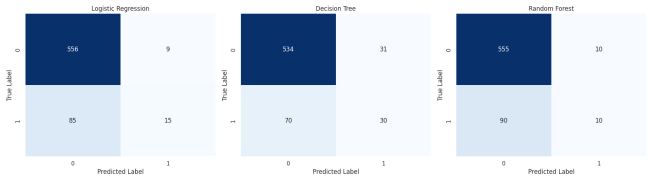


Fig. 7. Confusion Matrix of Task-1 Classification

all performance, achieving the highest prediction accuracy of 86% (shown as redTable 6) and the confusion matrix shown as redFigure 7. While maintaining favorable precision, recall, and F1-score balance across classes. Logistic regression was selected as the final predictive model for estimating engagement.

4.2 Task2 Analysis: Identifying Risk of Churn

The analysis and process of this task are similar to Task 1, both of which are **binary classification tasks**. The difference is that Task 2 is used to determine whether there is a risk of user churn, help the company find users who may be lost, and then help company make timely decisions to recover.

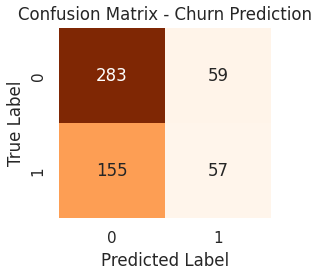


Fig. 8. Confusion Matrix for Task2 Churn Prediction

This confusion matrix (shown redFigure 8) and statistics (shown redTable 7) performs well in identifying customers who have not churned, but has significant deficiencies in identifying actual churned customers (155 missed customers). This means that the model fails to effectively identify potential churned customers. Therefore, the model urgently needs to improve its recall rate to more effectively support business decisions.

Table 7. Churn Prediction Report (Classification Metrics)

Class	Precision	Recall	F1-score	Support
0	0.65	0.83	0.73	342
1	0.49	0.27	0.35	212
Accuracy		0.61		554
Macro Avg	0.57	0.55	0.54	554
Weighted Avg	0.59	0.61	0.58	554

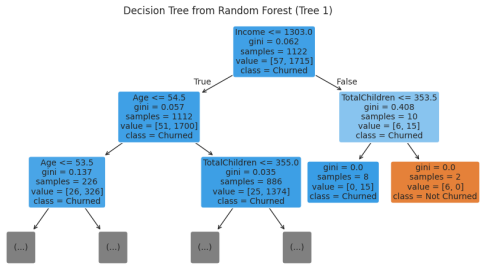


Fig. 9. Sub-Decision Tree of Task2 (No.1)

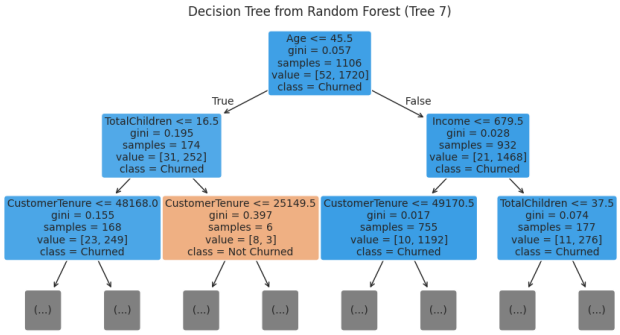


Fig. 10. Sub-Decision Tree of Task2 (No.7)

The above is just a random selection of two trees in the random forest for visualization, and the depth of the tree is limited. However, based on the above figure (shown redFigure 8) and table (shown redTable 7) analysis, we still need to improve the model's ability to judge "non-churned" users and use better data sets for centralized training.

4.3 Task3 Analysis: Forecast Purchase Wills

This task predicts consumer will purchase specific types of goodsfrom daily necessities (e.g.milk, meat, fruits) to luxury items (e.g.gold)based on features like family structure, income, consumption preferences, and marketing engagement. It is framed as a **multi-label binary classification**, where each product type is treated as an **independent binary task**. This approach captures the purchase intent across categories and the heterogeneous impact of features. The analysis structure aligns with Task 1, and three modelslogistic regression, decision tree, and random forestare used for comparison.

A quantitative table summarizes the prediction results. (shown as redFigure 11)

4.3.1 Visualization of Task3

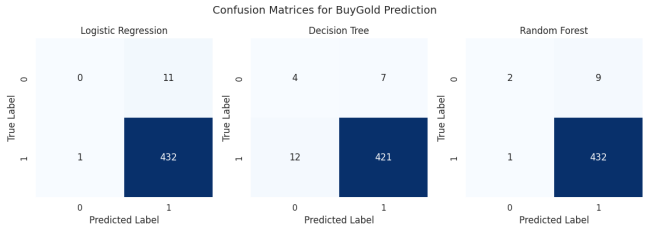


Fig. 11. Confusion Matrix of Task-3 Classification

Table 8. Task3 Classification Model Comparison

Indicator	LR		DT		RF	
	0	1	0	1	0	1
Precision	0.00	0.98	0.25	0.98	0.67	0.98
Recall	0.00	1.00	0.36	0.97	0.18	1.00
F1-Score	0.00	0.99	0.30	0.98	0.29	0.99
Accuracy		0.97		0.96		0.98
Macro F1		0.49		0.64		0.64
Weighted F1		0.96		0.96		0.97

The model's Precision, Recall, and F1-Score for identifying Class 0 (minority class) are significantly lower, indicating that the data is seriously imbalanced, with far fewer Class 0 samples than Class 1. Accuracy alone is misleading.

4.3.2 Future Work to Improve Task3

Future work will alleviate the imbalance problem by collecting more data (improving data quality and balance), adjusting class weights, and other data and model-level methods, and introduce PR curves, ROC curves and other indicators for comprehensive evaluation.

4.4 Task4 Analysis: Detect Low-Activity Guys

This task conducts an in-depth analysis of the user's **relation between browsing and purchasing behavior** on the website, focusing on the user characteristics of "frequent visits but no purchases". First, a scatter plot (shown as redFigure 12) is used to show the relationship between the number of monthly web page visits and the number of web page purchases by users, and colors are used to distinguish and reflect the relationship between activity and purchasing behavior. Subsequently, a bubble chart (shown as redFigure 12) is generated through group statistics. The size of the bubble represents the number of users under this combination, revealing the user group with high frequency of visits and low purchases.

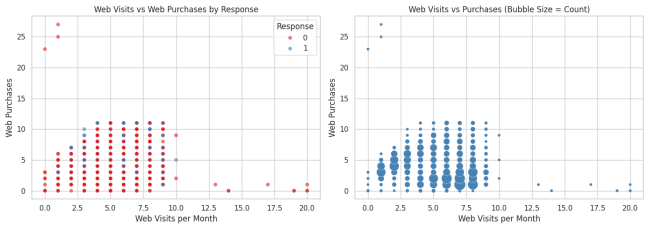


Fig. 12. Scatter(left) and Bubble Chart(right) Reflecting the Relationship between Web-Visits and Purchase

Next, heat maps (shown as redFigure 13) and three-dimensional bar charts (shown as redFigure 14) are used to show the density and structure of user distribution, enhancing the multi-dimensional visualization of the data. Finally, a box plot is used to compare the catalog purchase differences between responding and non-responding users, and the key indicators (number of visits, number of purchases, number of catalog purchases) are described by means, medians, etc.

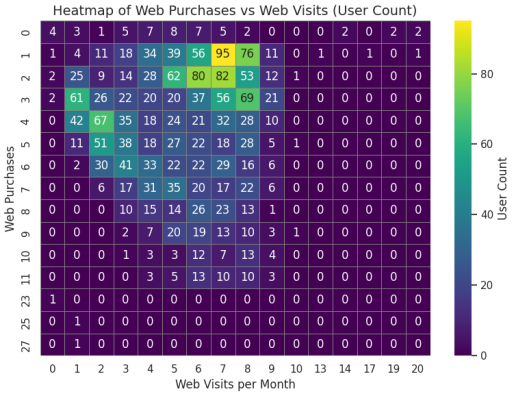


Fig. 13. Heatmap Reflecting the Relationship between Web-Visits and Purchase

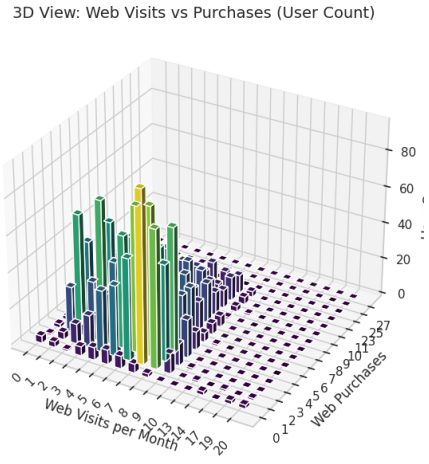


Fig. 14. 3D Visualization of Reflecting the Relationship between Web-Visits and Purchase

4.4.1 Problem Discovery

In addition, screening out users with high visit times (10 times/month) but no purchase behavior (shown as redTable 10) provides potential customer churn or conversion opportunities, providing data support for the subsequent formulation of precision marketing strategies.

Table 9. User Behavior Statistics by Response (Vertical)

Times (month)	Response = 0		Response = 1	
	Value	Idx	Value	Idx
WebVisits	5.32	Mean	5.31	Mean
	6	Median	6	Mid
	20	Max	10	Max
	0	Min	1	Min
WebPurchases	3.91	Mean	5.07	Mean
	3	Median	5	Mid
	27	Max	11	Max
	0	Min	0	Min
CatalogPurchases	2.40	Mean	4.20	Mean
	1	Median	4	Mid
	28	Max	11	Max
	0	Min	0	Min

Table 10. Users with Frequent Web Visits (10 per month) but No Web Purchases (Low-activity Users)

Index	ID	NumWebVisitsMonth	Response
774	4303	20	0
981	3955	14	0
1042	10749	19	0
1245	6862	20	0
1524	11110	14	0

4.4.2 Future Work to Solve Task4

- **Personalized Recommendations**: Analyze users browsing paths, time spent on pages, and preference tags to deliver product or service recommendations that better align with their interests.
- **Targeted Incentives**: Offer exclusive coupons, time-limited discounts, or free shipping policies to encourage initial purchases from these users.
- **A/B Test of Page Content or Layout**: Frequent visits without conversions may indicate low page appeal or unclear messaging. Test variations in product descriptions, image formats, and layout to enhance engagement.
- **User Intent Profiling**: Assess whether users are price comparison-oriented or information-seeking, and tailor marketing strategies accordingly.

4.5 Task 5&6 Analysis: Constructing a Value Evaluation Model to Profile High-Value Pair Customers and User Distribution Analysis

This task performs an **RFM (Recency, Frequency, Monetary)** analysis to identify high-value customers from a retail dataset. It first calculates total customer spending across various product categories, then assigns R, F, and M scores using quartile-based segmentation. Customers with an RFM score of 9 or higher are considered high-value. The script then visualizes their demographic and behavioral patterns using bar plots, scatter plots (shown as redFigure 15), and boxplots.

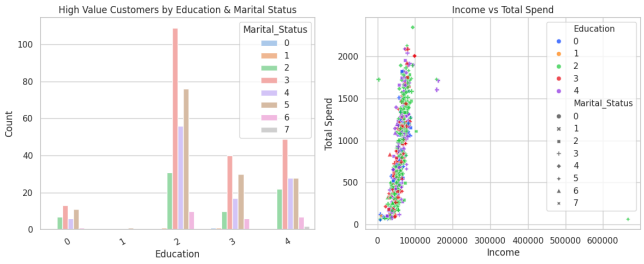


Fig. 15. High-Value Pair Customers Characteristics

4.5.1 RFM Model: Mathematical Principles

The RFM model evaluates customer value based on three key dimensions: Recency, Frequency, and Monetary value. Each dimension can be mathematically defined as follows:

- **Recency** (R_i): Measures how recently a customer made a purchase.

$$R_i = \text{CurrentDate} - \text{LastPurchaseDate}_i$$

Interpretation: A lower value, a more active.

- **Frequency** (F_i): Captures how often a customer makes purchases within specific time.

$$F_i = \sum_{t=1}^{T} \mathbf{1}_{\text{purchase}_{it}}$$

Interpretation: A higher value, a greater.

- **Monetary** (M_i): Represents the total monetary value of a customer's purchases during a given time period.

$$M_i = \sum_{t=1}^{T} \text{PurchaseAmount}_{it}$$

Interpretation: A higher value, a stronger power.

4.5.2 RFM Scoring Method

Each dimension is typically scored using either quantile-based bucketing or ranking. Scores usually range from 1 to 4 or 1 to 5.

* For **Recency**, lower values receive higher scores (more recent = better).
* For **Frequency** and **Monetary**, higher values receive higher scores (more frequent or higher spending = better).

After scoring, a composite RFM score is computed using either weighted summation or concatenation:

$$\text{RFM_Score}_i = R_i^* + F_i^* + M_i^*$$

where R_i^*, F_i^*, and M_i^* represent the normalized or ranked scores of each component and its visualization effects are shown as fellows (redFigure 18):

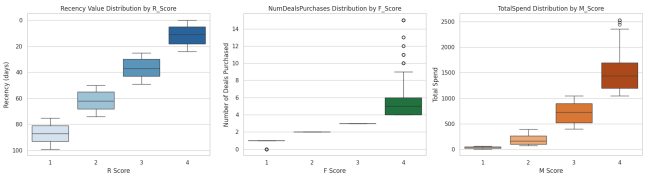


Fig. 16. Boxplot of RFM Indicator with R/F/M Score

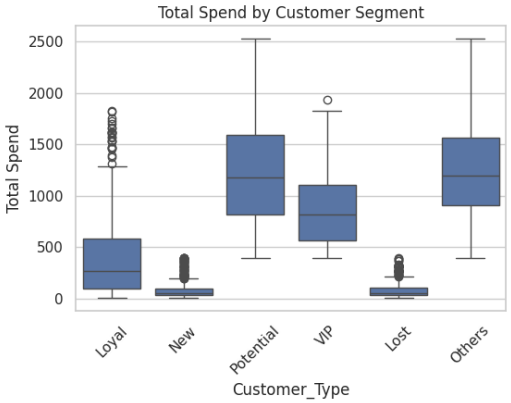


Fig. 17. Boxplot of Different Types of Customers

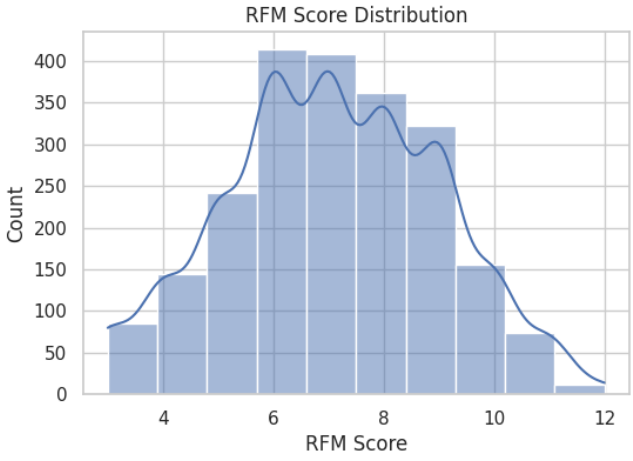


Fig. 18. RFM Score Distribution Plot



Fig. 19. Ability Evaluation Indicators of Different Types of Customers (Loyal VIP)



Fig. 20. Distribution of Different Types of Customers

4.5.3 Insights from the Distribution

* **Bell-Shaped Distribution:** The RFM scores mostly range from 3 to 12, with the peak between 6 and 8. This indicates a normal-like distribution, suggesting most customers are moderately active and valuable.
* **High-Value Tail:** A small portion of users score between 10 and 12, representing VIP or high-value customers.
* **Low-End Tail:** Customers with RFM scores between 3 and 5 are likely to be at risk of churn or exhibit low engagement.

4.5.4 Strategic Business Recommendations

* **Middle Segment (Scores 6–8):**
  - Nurture this group to increase customer value.
  - Offer personalized incentives such as loyalty points or targeted cross-selling.
* **High RFM Segment (Scores 9–12):**
  - Focus on retention and recognition.
  - Implement VIP programs, provide exclusive access to products or promotions, and gather feedback.
* **Low RFM Segment (Scores 3–5):**
  - Target with re-engagement campaigns.
  - Use welcome-back discounts or evaluate ROI before continued targeting.

* **A/B Testing Recommendations:**
  - Conduct A/B testing on lowmid RFM segments using different messaging, pricing, or interface strategies to encourage activity.

The RFM score distribution reflects a healthy middle-ground customer base. Businesses can:

* Upsell to the middle segment,
* Retain and reward high-value users,
* Reactivate or filter out low-value users.

This segmentation supports a data-driven customer relationship strategy aimed at optimizing marketing ROI.

5 External Analysis

5.1 Customer Segmentation and SWOT Strategic Analysis for Market Insights

The age distribution of the company's customers is primarily concentrated in the middle-aged and elderly group, ranging from 35 to 70 years old (shown as redFigure 5). To gain deeper insights into consumer behavior, we categorized users into six segments: **Loyal**, **Others**, **New**, **Potential**, **Lost**, and **VIP** (shown as redFigure 20). Each of these segments exhibits distinct consumption habits, purchasing power, income levels, and responsiveness to marketing activities (shown as redFigure 19).

Following this segmentation, we analyzed the distribution of users involved in the companys sales activities (shown as redFigure 20). On the one hand, the proportion of loyal users is relatively high, indicating strong customer stickiness and brand loyalty. Additionally, the presence of new and potential users suggests ongoing market interest and growth potential. On the other hand, there are some concerning signals: the number of **VIP users** who are typically expected to be high spendersis relatively small, and more critically, their actual contribution to revenue growth has not shown a significant increase.

Taking into account multiple indicators such as consumption power, income level, and responsiveness to campaigns, we find a large proportion of users cencerted on **middle tier** of the comprehensive user scoring system (see Table red18). Furthermore, our study has identified subgroups of **low-engagement users** (see Table red10) and **high-value users** (see at **Appendix**). These segments provide opportunities for targeted promotions and personalized marketing strategies.

From a strategic standpoint, this segmentation provides valuable insights for a broader:

5.2 SWOT analysis

* **Opportunities:**
  - The large base of potential and new users presents a strong opportunity for future conversion through engagement campaigns and loyalty programs.
  - Identifying high-value users enables the design of **premium service** offerings and personalized product recommendations to maximize revenue.
* **Threats:**
  - The low number and underperformance of VIP users may indicate strong market competition or **ineffective high-end user engagement strategies.**
  - The high proportion of mid-tier and low-engagement users highlights a risk of stagnation if customer experience and product offerings are not innovated.

These findings underscore the importance of refining customer relationship management strategies, leveraging data-driven segmentation to both retain loyal users and activate underperforming segments.

6 Options and Strategic Alternatives

To address the weak contribution of VIP customers and unlock their potential value, the company may consider the following strategic alternatives:

* **Tiered Loyalty Program Enhance**
  Execution: Redesign the existing loyalty program with multiple tiers (e.g., Silver, Gold, Platinum), each offering exclusive benefits such as priority access, higher discounts, and early product launches.
  Resources Required: CRM platform upgrades, personalized marketing content, customer data enrichment.
  Expected Impact: Increase in VIP customer engagement and purchase frequency, improved customer lifetime value (CLV).
  Cost-Benefit Analysis:
  - **Costs:** Medium to High (system upgrades, marketing content development)
  - **Benefits:** High (boosted retention, higher AOV, brand differentiation)

* **High-Value Concierge Service**
  Execution: Establish a VIP concierge or relationship manager system offering 1-on-1

assistance, customized product recommendations, and private sales events.
Resources Required: Dedicated service staff, training, CRM workflow integration.
Expected Impact: Strengthened brand loyalty and customer satisfaction; creates a luxury brand image.
SWOT Analysis:

- **Strength:** Personalized experience increases perceived value.
- **Weakness:** High operational cost and limited scalability.
- **Opportunity:** Potential to convert high-spenders into long-term loyalists.
- **Threat:** May alienate non-VIP customers if perceived as exclusive.

* **Behavior-Based VIP Identification and Dynamic Targeting**
Execution: Use machine learning models to dynamically identify potential high-value users based on browsing, purchasing, and referral behaviors, then serve them targeted offers and onboarding experiences.
Resources Required: Data science team, model training infrastructure, integration with marketing automation tools.
Expected Impact: Expands the VIP segment intelligently and early; improves campaign ROI by focusing on high-propensity users.
Cost-Benefit Analysis:
  - **Costs:** Medium (initial data infrastructure and modeling)
  - **Benefits:** High (efficient targeting, reduced marketing waste)

A hybrid strategycombining loyalty tier refinement with behavior-based dynamic targetingoffers a scalable, data-driven approach to strengthen VIP engagement while minimizing resource waste. Choosing the appropriate strategy will depend on the companys technical readiness and long-term growth objectives.

7 Conclusion

In this study, we conducted a comprehensive analysis of customer segmentation using **RFM modeling**, focusing on demographic features such as age, as well as behavioral data including recency, frequency, and monetary spending. The majority of consumers were found to fall within the 3570 age range, indicating a concentration in the **middle-aged** and senior demographic segment. By categorizing customers into six tiers—***Loyal***, ***Others***, ***New***, ***Potential***, ***Lost***, **and** ***VIP***—we were able to better understand their distinct purchasing patterns, income capabilities, and responsiveness to promotional efforts.

Our segmentation results revealed a relatively high proportion of loyal and potential customers, suggesting that the company's product offerings exhibit strong stickiness and customer retention. However, the analysis also highlighted critical challenges: despite their potential, VIP customers are few in number and their contribution to overall revenue growth is not proportionately significant.

Based on these insights, we identified both **opportunities and threats**, particularly regarding the under-leveraged high-value segments and the presence of low-engagement customers.

The findings underscore the importance of data-driven customer management strategies in a competitive landscape with rising market saturation and evolving consumer behavior. In **future work,** we recommend implementing longitudinal tracking of user behavior to dynamically adjust segmentation models and validate the long-term effectiveness of the proposed strategic alternatives.

8 Summary

Our analysis reveals that the majority of consumers are aged between 35 and 70, with strong representation from Loyal and Potential customer segments. While Loyal customers demonstrate high retention and purchasing consistency, VIP customersdespite their expected contributiondo not show significant impact on revenue growth. The study also identifies a substantial portion of users with moderate composite scores.

Based on these findings, we recommend implementing tailored engagement strategies, investing in converting Potential users into Loyal ones, and reevaluating the resource allocation for VIP users. This customer-centric approach enables the company to enhance lifetime value, increase campaign effectiveness, and maintain a competitive edge in a saturated market.

9 Acknowledgment

The authors gratefully acknowledge Dr. Omar Romero-Hernandez and the Berkeley Haas School of Business for providing access to this valuable dataset, which served as the foundation for our analysis. We also appreciate the support and guidance received throughout the project, which greatly contributed to the quality and depth of this research.

10 APPRNDIX

10.1 Customer Segmentation via PCA Projection and K-Means Clustering

In order to uncover meaningful behavioral patterns and purchasing preferences among customers, we conduct a segmentation analysis using the **K-Means clustering algorithm** on selected features. The selected features include:

* Age
* Income
* TotalSpend
* Recency
* CustomerTenure

10.2 Feature Scaling

To ensure all variables contribute equally to the clustering process, we apply standard normalization:

$$\tilde{x}_{ij} = \frac{x_{ij} - \mu_j}{\sigma_j}, \quad j = 1, 2, \ldots, p$$

where μ_j and σ_j denote the mean and standard deviation of feature j, respectively.

10.3 Clustering with K-Means

We apply the K-Means algorithm with $K = 4$ clusters to the standardized dataset. The objective function of K-Means is defined as:

$$object\ func = \arg\min_{S} \sum_{k=1}^{K} \sum_{x_i \in S_k} \|x_i - \mu_k\|^2$$

where μ_k is the centroid of cluster S_k. The result is the assignment of each customer to one of four segments based on proximity to centroids in the feature space.

10.4 Dimensionality Reduction and Visualization

To visualize customer clusters in a two-dimensional space, we perform Principal Component Analysis (PCA). PCA seeks to find the orthogonal linear transformation that maximizes variance. The transformed components are defined by:

$$Z = XW, \quad W^\top W = I$$

where $X \in R^{n \times p}$ is the standardized data matrix and $W \in R^{p \times 2}$ are the top two eigenvectors corresponding to the largest eigenvalues of the covariance matrix $X^\top X$.
The PCA plot visually separates customer segments, and the red centroids in the PCA projection highlight the cluster centers derived from K-Means.

10.5 Segment Profiling

For each segment, we calculate the mean value of each input feature:

$$\mu_{k,j} = \frac{1}{|S_k|} \sum_{x_i \in S_k} x_{ij}, \quad j = 1, \ldots, p$$

This profiling allows us to understand the demographic and behavioral differences between customer groups. We visualize these differences through:

10.6 Visualization

* Boxplots (shown as redFigure 21) of individual feature distributions across clusters
* Radar charts shown as redFigure 22(normalized via MinMax scaling to $[0,1]$) for feature comparison
* Line charts (shown as redFigure 22) of cluster-wise feature means

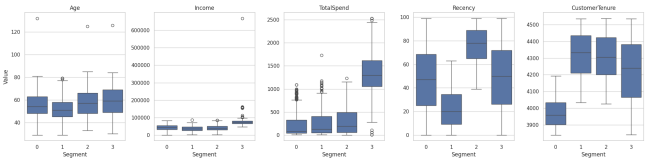


Fig. 21. Boxplot of Five Key Features Corresponding to Four Clusters

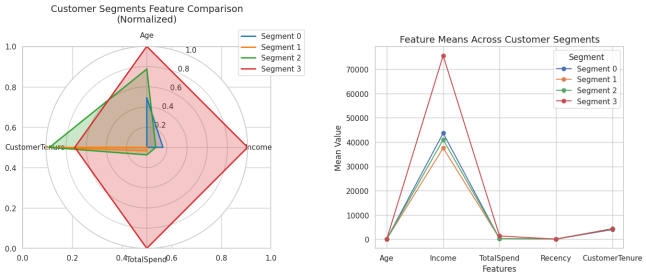


Fig. 22. The Capabilities of Four Clusters Divided by PCA

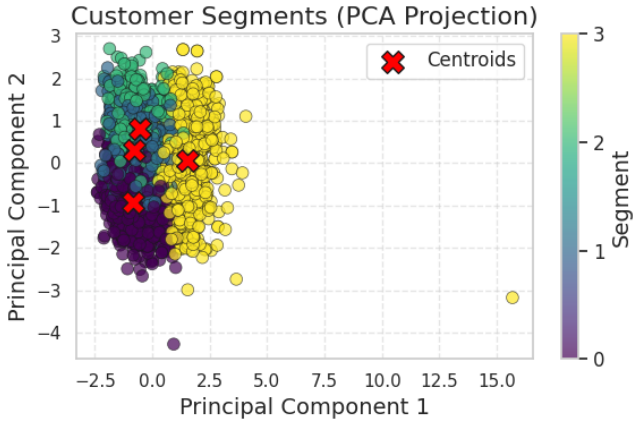


Fig. 23. PCA Customer Segementation

10.7 High-Value Customer Table

Next, based on the above comprehensive RFM statistical scoring method and the dual screening of the clustering algorithm, we discovered that there are 410 high-value users in the consumer group. Next, we will show 10 of them. (shown as redTable 11)

Table 11. Selected Features from High-Value Customers Dataset (selected 10 of total 410 samples)

ID	TotalSpend	RFM_Score	Recency	Income
2114	1315	9	19	47699
6565	1782	7	76	78649
1966	1672	6	96	75332
8755	1196	9	11	44370
8601	1395	10	16	56017
6566	1319	11	16	69239
4114	1693	9	41	61622
2225	1617	7	59	70482
6260	1957	7	47	80313
10383	1093	8	17	46498

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